

CH4203: Organic Synthesis Laboratory

Expt. No. 1: Nitro-Aldol Reaction of *p*-Tolualdehyde with Nitromethane: Synthesis of β -Nitrostyrene

Aim and Objectives: In this lab, we will learn **Henry Reaction** which is nothing but a **nitro-aldol** reaction and we will make a substituted β -nitro styrene using nitroaldol reaction followed by elimination.

Apparatus Requirement:

Round Bottom (RB) flask; Conical flask; Measuring cylinder; Beaker; Separating funnel; Celite funnel; Filter paper.

Chemicals Requirement:

1. *p*-tolualdehyde
2. Nitromethane
3. Sodium Hydroxide
4. Hydrochloric Acid
5. Methanol
6. Ethanol

Molecular Wt. and Physical Details of Reagents:

Molecular formula of 4-Methyl Benzaldehyde = C_8H_8O

Molecular weight of 4-Methyl Benzaldehyde = 120.15 g/mole

Molecular formula of Nitromethane = CH_3NO_2

Molecular weight of Maleic Anhydride = 61.04 g/mole

Experimental Procedure:

In a round-bottom flask fitted with a stirrer, 5 mL (42.4 mmol) of *p*-tolualdehyde, 3.4 mL (63.6 mmol) of nitromethane ($MeNO_2$) were taken in 50 mL of MeOH.

A solution of sodium hydroxide is prepared by dissolving 2.54 g (63.6 mmol) of NaOH in approximately 10 mL of water and was added to the reaction mixture at 0 °C. A bulky white precipitate forms rapidly during the addition of the alkali. The mixture gets so thick that stirring becomes difficult and it may be advisable to add 25 mL more of MeOH.

The reaction was slowly warmed to room temperature and stirred for 30 minutes at room temperature.

After stirring of 30 minutes, the pasty mass is converted to a clear solution by the addition of 5 mL of ice water containing crushed ice.

Next, 50 mL hydrochloric acid (6N HCl) was taken in a beaker and cooled it to 0 °C. Later, the reaction mixture was slowly added to this cold solution.

A pale-yellow crystalline mass separates almost immediately as the alkaline solution comes in contact with the acid and the solid settles to the bottom of the beaker.

The major part of the cloudy liquid layer is removed by decantation, and the residue filtered by suction and washed with water until free from chlorides.

The crude β -nitrostyrene then purified by the re-crystallization in 50 mL of hot EtOH. The yield of recrystallized β -nitrostyrene is approximately 80-85%.

Melting Point: 57 – 59 °C

R_f (20%EtOAc in *n*-hexane) = 0.39

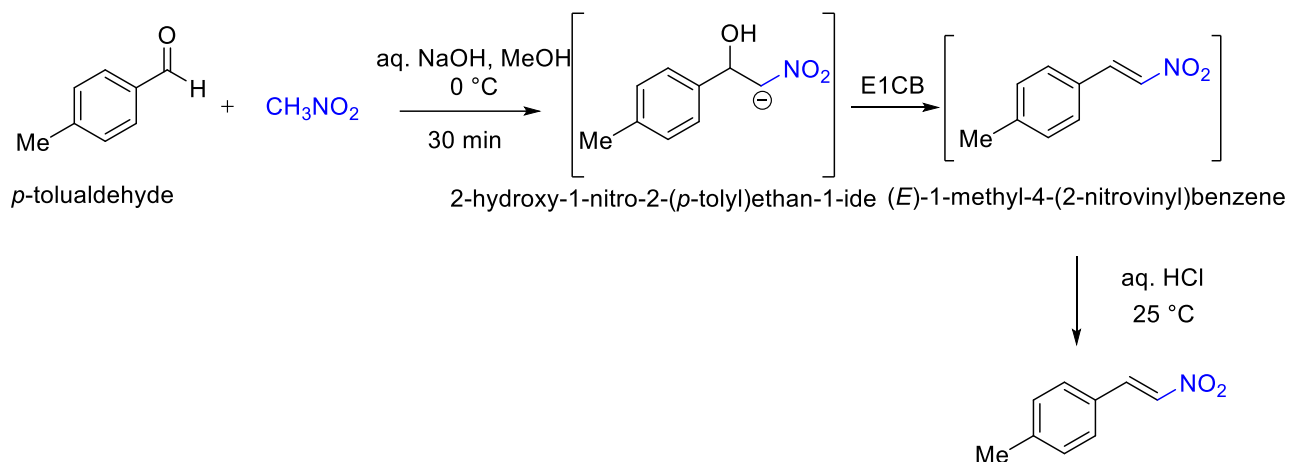
Note: The vapour of hot solution of β -nitrostyrene is very irritating to the eyes and nose, and the skin of the face is sensitive to the solid substance.

Characterization of β -nitrostyrene:

In summary, we have conducted the nitro-Aldol reaction of *p*-tolualdehyde with nitromethane followed by the elimination for the synthesis of β -nitrostyrene.

The final product was characterized by **IR** spectra, **¹H NMR** spectra, **¹³C NMR** spectra, and **HRMS** spectroscopy.

Reaction Mechanism:



Results & Observation:

Yield: 5.8 g

% of Yield: 84%

MP: 57 – 59 °C

TLC System: $R_f = 0.39$ (20% EtOAc in *n*-hexane)

Compound Characterization:

IR Frequencies: 3152, 1629, 1557, 1268 cm^{-1}

^1H -NMR: (500 MHz, CDCl_3) δ 8.00 (d, $J = 13.7, 2.2$ Hz, 1H), 7.59 (d, $J = 13.7, 2.0$ Hz, 1H), 7.49 – 7.44 (d, 2H), 7.28 (d, $J = 7.7$ Hz, 2H), 2.43 (s, 3H).

^{13}C -NMR: (125 MHz, CDCl_3) δ 143.1, 139.2, 136.3, 130.2, 129.2, 127.3, 21.7

Conclusion:

In summary, we have done nitro-aldol reaction of p-tolualdehyde with nitromethane to synthesize β -Nitrostyrene. The chemical yield was 84%. Further, the compound was characterised by IR spectra, NMR spectra and melting point was also checked.